Lecture 29
Matched pairs; categories

Review

1 What does $df = \min(n_1 - 1, n_2 - 1)$ approximate?

the Welch df that software uses: a smaller df, so a conservative P-value

2 When do we pool for two means?

only when $\sigma_1 = \sigma_2$ is reasonable (similar SDs); unpooled is our default

3 What is the pooled SD s_p?

one SD from both samples: $s_p^2 = ((n_1 - 1)s_1^2 + (n_2 - 1)s_2^2)/(n_1 + n_2 - 2)$

The rank-sum test (non-parametric)

Wilcoxon rank-sum test

compares two independent samples without assuming normal populations

how it works

rank all $n_1 + n_2$ values together, then compare the two samples' rank sums

when to use it

small samples from skewed populations, outliers, or ranked (ordinal) data

more

Resources → Extras → Non-parametric Tests (also the signed-rank test for pairs)

Matched pairs: the idea

matched pairs

each value in one sample is paired with one in the other: same subject twice

the differences

$d = x_1 - x_2$ **for each pair: the n differences are the data**

the parameter

μ_d: **the mean difference in the population of pairs**

the big idea

a two-sample design becomes a one-sample t-test on the differences

The matched-pairs t-test

$H_0: \mu_d = 0$

no difference within pairs, on average

H_A

$\mu_d > 0$ **(upper),** $\mu_d < 0$ **(lower),** $\mu_d \neq 0$ **(two-sided)**

the test statistic

$t_{obs} = (\bar{d} - 0)/(s_d/\sqrt{n})$: $\bar{d}, s_d$ **= the mean and SD of the differences**

degrees of freedom

$df = n - 1$, **where n is the number of pairs**

assumptions

a random sample of pairs; the differences are normal, or $n \geq 30$

Data: hypnotism and pain

Subject	A	B	C	D	E	F	G	H
Before	6.6	6.5	9.0	10.3	11.3	8.1	6.3	11.6
After	6.8	2.4	7.4	8.5	8.1	6.1	3.4	2.0
d = after − before	0.2	−4.1	−1.6	−1.8	−3.2	−2.0	−2.9	−9.6

Randomly selected subjects had a sensory pain score measured before and after hypnotism; a lower score means less pain. X = a subject's score before, Y = the same subject's score after, $d = Y - X$. $n = 8$ pairs. (OpenStax, Example 10.11)

Example: does hypnotism reduce pain?

A study asked whether hypnotism reduces pain. Each subject's pain score was measured before and after hypnotism (lower means less pain), and the differences are normal. Are the scores lower after hypnotism, on average? (OpenStax, Example 10.11)

$n = 8$ **subjects;** d **= after – before;** $\bar{d} = -3.125,\ s_d = 2.91$**;** $\alpha = 0.05$

(a) Why matched pairs? State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

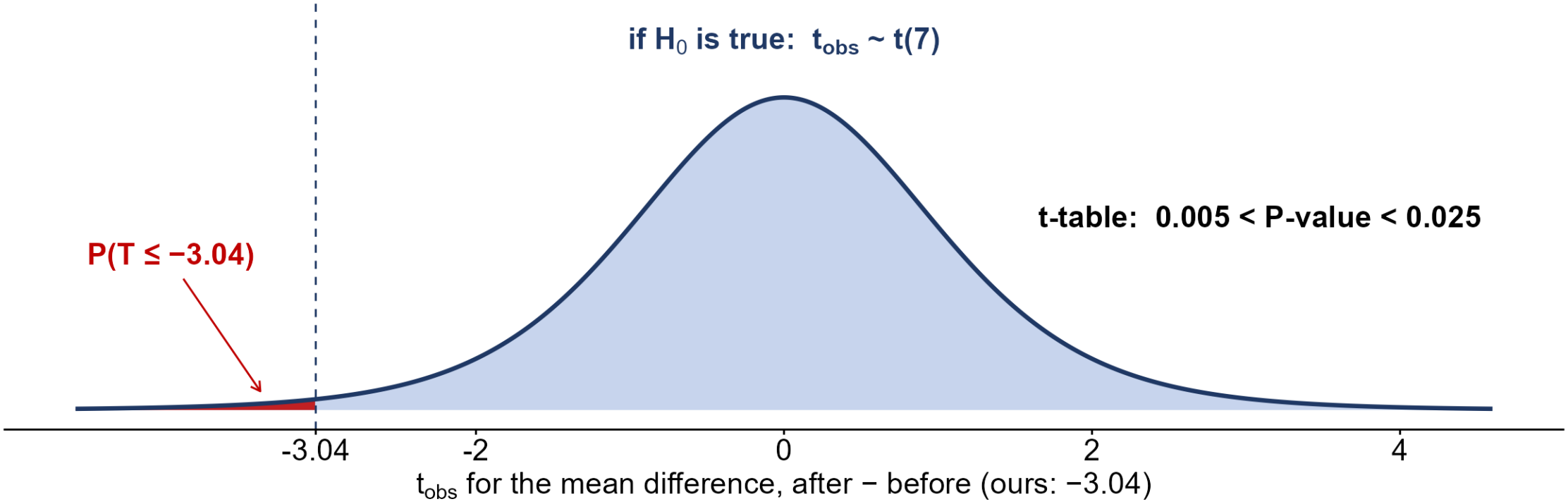
(c) Find the P-value from the t-table. What is the decision?

(a) the same 8 subjects, twice; $H_0: \mu_d = 0$ $H_A: \mu_d < 0$ **(lower-tailed)**

(b) $\text{SE} = 2.91/\sqrt{8} = 1.0288;\ t_{obs} = -3.125/1.0288 = -3.04;\ df = 8 - 1 = 7$

(c) $2.365 < 3.04 < 3.499$**:** $0.005 <$ **P-value** < 0.025**, below 0.05: reject** H_0

The hypnotism P-value



$0.005 <$ **P-value** < 0.025 **(software: 0.0094): pain scores are lower after hypnotism, on average**

Why pair?

Suppose we forgot the pairing and treated the hypnotism scores as two independent samples of 8.

before: $\bar{x} = 8.71,\ s = 2.18$; **after:** $\bar{x} = 5.59,\ s = 2.61$; $\bar{x}_{\text{after}} - \bar{x}_{\text{before}} = -3.125$

(a) Find the unpooled SE and t_{obs}.

(b) Compare with the matched-pairs test. Why is the paired SE smaller?

(a) $\text{SE} = \sqrt{2.18^2/8 + 2.61^2/8} = 1.2023;\ \ t_{obs} = -3.125/1.2023 = -2.60$

(b) paired $\text{SE} = 1.0288 < 1.2023$**: each subject is their own control**

Data: shot-put with each hand

Student	1	2	3	4	5	6	7
Dominant hand	30	26	34	17	19	26	20
Weaker hand	28	14	27	18	17	26	16
d = dominant − weaker	2	12	7	−1	2	0	4

Seven eighth graders pushed the shot-put with their dominant (writing) hand and with their weaker hand, in feet. X = the dominant-hand distance, Y = the weaker-hand distance, $d = X - Y$. $n = 7$ pairs. (OpenStax, Example 10.13)

Practice: shot-put with each hand

Seven eighth graders thought they could push the shot-put equal distances with either hand. Each pushed with their dominant hand and with their weaker hand; the differences are normal. Is there a mean difference between the two hands? (OpenStax, Example 10.13)

$n = 7$ **students;** d **= dominant – weaker (feet);** $\bar{d} = 3.71,\ s_d = 4.50;$ $\alpha = 0.05$

(a) Why matched pairs? State H_0 and H_A. Which tail?

(b) Find the SE and t_{obs}. What is the df?

(c) Find the P-value from the t-table. What is the decision?

(a) the same 7 students, both hands; $H_0: \mu_d = 0$ $H_A: \mu_d \neq 0$ **(two-sided)**

(b) $\text{SE} = 4.50/\sqrt{7} = 1.7008;$ $t_{obs} = 3.71/1.7008 = 2.18;$ $df = 7 - 1 = 6$

(c) $1.943 < 2.18 < 2.447:$ $0.05 <$ **P-value** < 0.10**, above 0.05: do not reject** H_0

Practice: which test?

1 **Mean nightly sleep: random samples of 40 first-years and 40 seniors**

two independent samples t-test: two separate samples, a quantitative response

2 **Reaction time of 25 drivers, rested and after a sleepless night**

matched-pairs t-test: the same 25 drivers, measured twice

3 **Share who exercise daily: random samples of 200 dog and 200 cat owners**

two-proportion z-test: two independent samples, a yes/no response

Summary: every test so far

test (samples)	test statistic	SE, unpooled	SE, pooled
one proportion z (one sample)	$z_{obs} = \frac{\hat{p} - p_0}{\text{SE}}$	$\sqrt{\frac{p_0(1-p_0)}{n}}$	–
one mean t (one sample)	$t_{obs} = \frac{\overline{x} - \mu_0}{\text{SE}}$	$\frac{s}{\sqrt{n}}$	–
two proportions z (independent)	$z_{obs} = \frac{(\hat{p}_1 - \hat{p}_2) - d_0}{\text{SE}}$	$\sqrt{\frac{\hat{p}_1(1-\hat{p}_1)}{n_1} + \frac{\hat{p}_2(1-\hat{p}_2)}{n_2}}$	$\sqrt{\hat{p}(1-\hat{p})\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$
two means t (independent)	$t_{obs} = \frac{\overline{x}_1 - \overline{x}_2}{\text{SE}}$	$\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$	$s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$
matched pairs t (dependent)	$t_{obs} = \frac{\overline{d} - 0}{\text{SE}}$	$\frac{s_d}{\sqrt{n}}$	–

pool only for $H_0: p_1 = p_2$, or when $\sigma_1 = \sigma_2$ is reasonable. Non-parametric: the sign test (pairs), the rank-sum test (independent)

Data: teens and phones

Answer	Never	Rarely	Sometimes	Often	Total
Count	289	216	178	60	743
Proportion	0.39	0.29	0.24	0.08	1.00

A Pew Research Center survey asked 743 U.S. teens aged 13 to 17: "Are you losing focus in class by checking your cell phone?" (Week 2)

X = a teen's answer: Never, Rarely, Sometimes or Often. $n = 743$.

More than two categories?

So far a categorical response had two outcomes (yes or no), and we tested one p or $p_1 - p_2$. The Pew question has four answers, and someone claims all four are equally common among U.S. teens.

Never 289, Rarely 216, Sometimes 178, Often 60; $n = 743$

(a) What kind of variable is X? How many categories does it have?

(b) Can a test of one proportion p check the claim? Why or why not?

(a) categorical, with $k = 4$ categories

(b) no: one p covers one category; the claim is about all four, p_1, p_2, p_3, p_4, at once

Tables for categorical data

one-way table

the count in each category of one categorical variable

two-way (contingency) table

the count for every combination of two categorical variables: rows × columns

marginal distribution

one variable on its own: the row or column totals, as proportions

conditional distribution

one variable within a single row (or column) of the other, as proportions

Data: volunteers and hours

Type of volunteer	1–3 hours	4–6 hours	7–9 hours	Row total
Community college	111	96	48	255
Four-year college	96	133	61	290
Nonstudents	91	150	53	294
Column total	298	379	162	839

Adults 21 and older volunteer one to nine hours a week with a disabled senior citizen. A sample of 839 volunteers: X = type of volunteer, Y = hours volunteered per week. (OpenStax, Example 11.6)

Example: do the rows look alike?

Is the number of hours volunteered related to the type of volunteer? Before any test, compare the rows of the two-way table.

1–3 hours: community college 111 of 255; nonstudents 91 of 294; everyone 298 of 839

(a) What share of community college volunteers work 1–3 hours? Of nonstudents?

(b) What share of all 839 volunteers work 1–3 hours? Which kind of distribution is each?

(c) If hours had nothing to do with type, what would the rows look like?

(a) $111/255 = 0.435$ **and** $91/294 = 0.310$**: conditional distributions**

(b) $298/839 = 0.355$**: the marginal distribution of hours**

(c) every row would have about the same split as the column totals: 0.355, 0.452, 0.193

Observed and expected counts

observed count O

the count in a category (or cell) of our data

expected count E

the count we would expect if H_0 is true: $E = n\,p_i$, with p_i from H_0

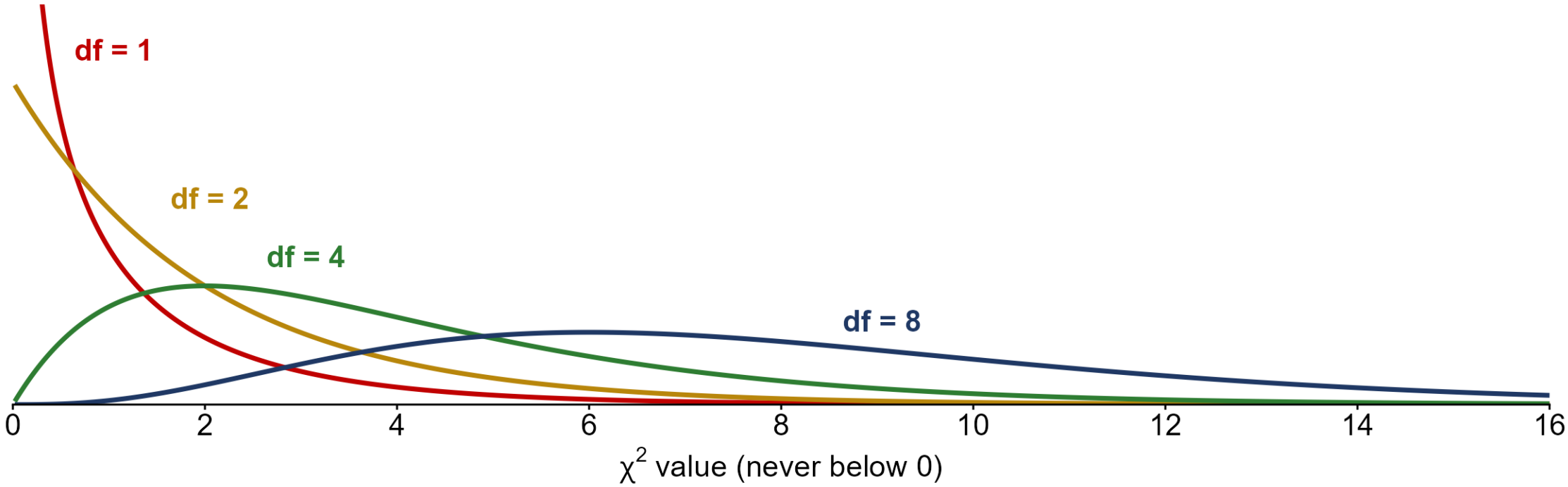
the chi-square distribution

a family of right-skewed curves for values ≥ 0, one for each df

degrees of freedom (one variable)

$df = k - 1$, **where k is the number of categories**

The chi-square distribution



never below 0, skewed right; mean $= df$; a larger df moves the curve right

The chi-square statistic

the chi-square statistic

$$X^2 = \sum \frac{(O - E)^2}{E}$$

added up over every category: how far each count is from what H_0 predicts

$X^2 = 0$

the observed counts match H_0 exactly

a large X^2

the counts are far from H_0: evidence against H_0 (always right-tailed)

if H_0 is true

X^2 is approximately $\chi^2(k-1)$, when every $E \geq 5$

Example: equally common answers?

Back to the Pew survey. Someone claims the four answers are equally common among U.S. teens:

$H_0: p_1 = p_2 = p_3 = p_4 = 0.25$

$n = 743$; **observed: Never 289, Rarely 216, Sometimes 178, Often 60**

(a) Find the expected count E for each answer.

(b) Find X^2.

(c) What is the df? Is X^2 large for that chi-square curve?

(a) $E = 743 \times 0.25 = 185.75$ **for every answer**

(b) $X^2 = 57.39 + 4.93 + 0.32 + 85.13 = 147.77$

(c) $df = 4 - 1 = 3$; **the mean of $\chi^2(3)$ is 3, so 147.77 is far out in the right tail**